Functional description of Diagnosis

Boot Block

For

ABS10.3MBML

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Change list:

No.	Date	Author	Description
1	06/05/2019	Naina Naganath	Final Version – Created for ABS10.3MBML
2	01/06/2020	Nguyen Phuong	Minor change in Service $2E - Write Data by Identifier: BBNumber number of bytes correction

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1. Short Description

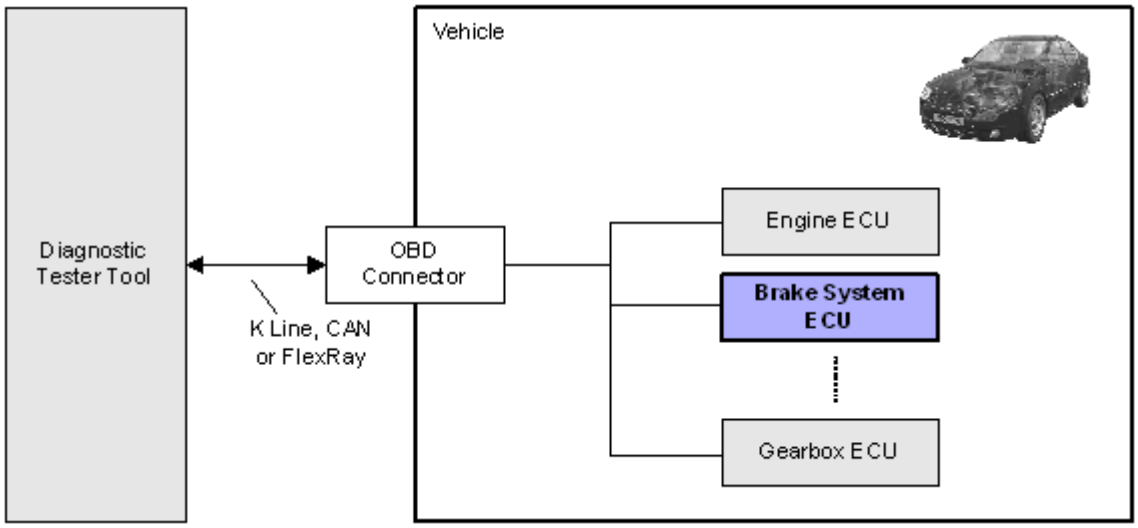
The boot block which is used for Reprogramming Function provides the capability of exchanging the application software of the control unit via external tester tools by means of diagnostic

1. Identify the ECU software
2. Unlock the ECU software security mechanism for unauthorized access
3. Erase the existing application software
4. Download a new application software.
5. Verify the authenticity of the downloaded data

Diagnostic boot block communication can be used in the servicing workshop. A safe and secure Reprogramming can be provided by the bootloader function with boot block data.

1.1 Operational Environment

Within a vehicle, normally several ECUs exist. These ECUs usually share a common diagnostic bus. For Boot block Diagnostic Communication a Tester is attachable to the Bus to perform Diagnostic and Download Communication.



(Example for Brake System ECU)

1.2 Use Cases

The Diagnostic Tester Tool can be a production facility in the vehicle production plant or a tool that is used in the repair shop. These tester tools send diagnostic requests to the ECUs in the vehicle and evaluate the responses. As the brake system ECU is also part of the vehicle, all Bosch brake system ECU's support a diagnosis communication interface to transfer information from an ECU to an off-board tester and vice-versa.

The information transferred consists of ECU fault information, signal reporting, control routines, calibration values and many other types of data used either in performing ECU diagnostics, re-programming or any other data transfer not associated with normal node to node communication.
The Bootblock as part of Diagnostic Communication has three major tasks:
1. Startup Consistency check, to evaluate if the application is valid and remain in secure boot mode if it is not.
2. Provide a safe reprogramming functionality, to exchange the application with different versions.

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3. Manage the history of all downloaded SW-versions and types.

2 Referenced documents

2.1 International documents

The general requirements of the diagnosis communication protocol are based on ISO requirements, specified in the following documents:

Reference	Title	Version
ISO 14229-1	Road vehicles —Unified diagnostic services (UDS) Part 1: Specification and requirements	ISO 14229-1:2013
ISO 14229-2	Road vehicles —Unified diagnostic services(UDS) Part 2: Session layer services	ISO 14229-2:2013
ISO 14229-3	Road vehicles —Unified diagnostic services (UDS) Part 3: Unified diagnostic services on CAN implementation (UDS on CAN)	ISO 14229-3:2012
ISO 15765-1 ISO 15765-2 ISO 15765-4	Road vehicles —Diagnostic communication over Controller Area Network (Do CAN)	ISO 15765-1:2011 ISO 15765-2:2011 ISO 15765-4:2011

AUTOSAR Specification of CAN Transport Layer, V2.1.1.

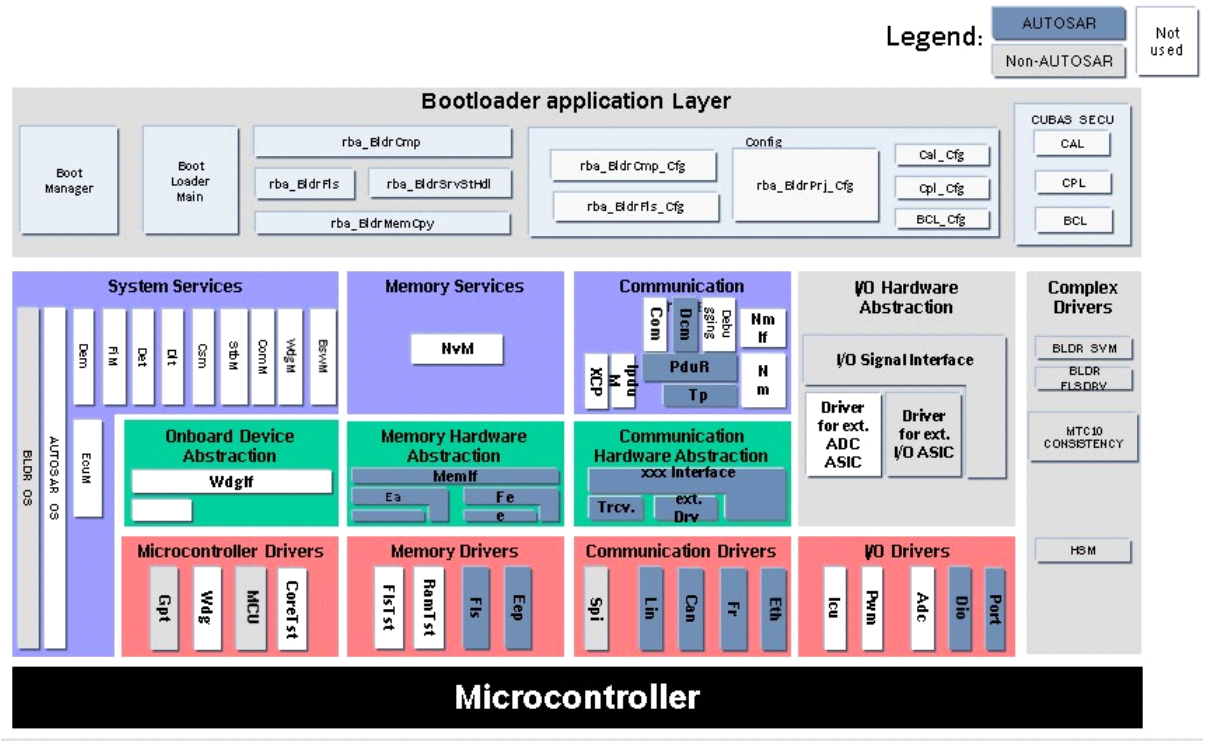
2.2 Abbreviations & Definitions

Abbreviation	Description
Bootblock	Independent Software which is permanently present in the ECU containing startup and SW-download functionality.
BootManager	Part of Bootblock, which performs SW-validity checks during start-up and starts either BootLoader or application SW.
BootLoader	Part of the Bootblock, which holds all functionality to erase, download and check application SW.
ECU	Electronic Control Unit; In this document the word ECU is used for the device the bootblock running on.
Tester	External tool, which requests service execution on the ECU. Tester - ECU communication is a Client - Server relation.
UDS	Unified Diagnostic Services; Specified diagnostic protocol in ISO14429
NRC	Negative Response Code; if a Service cannot be executed, the ECU responds with a negative response. The NRC is used to indicate the cause of the error.

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3 System Behaviour

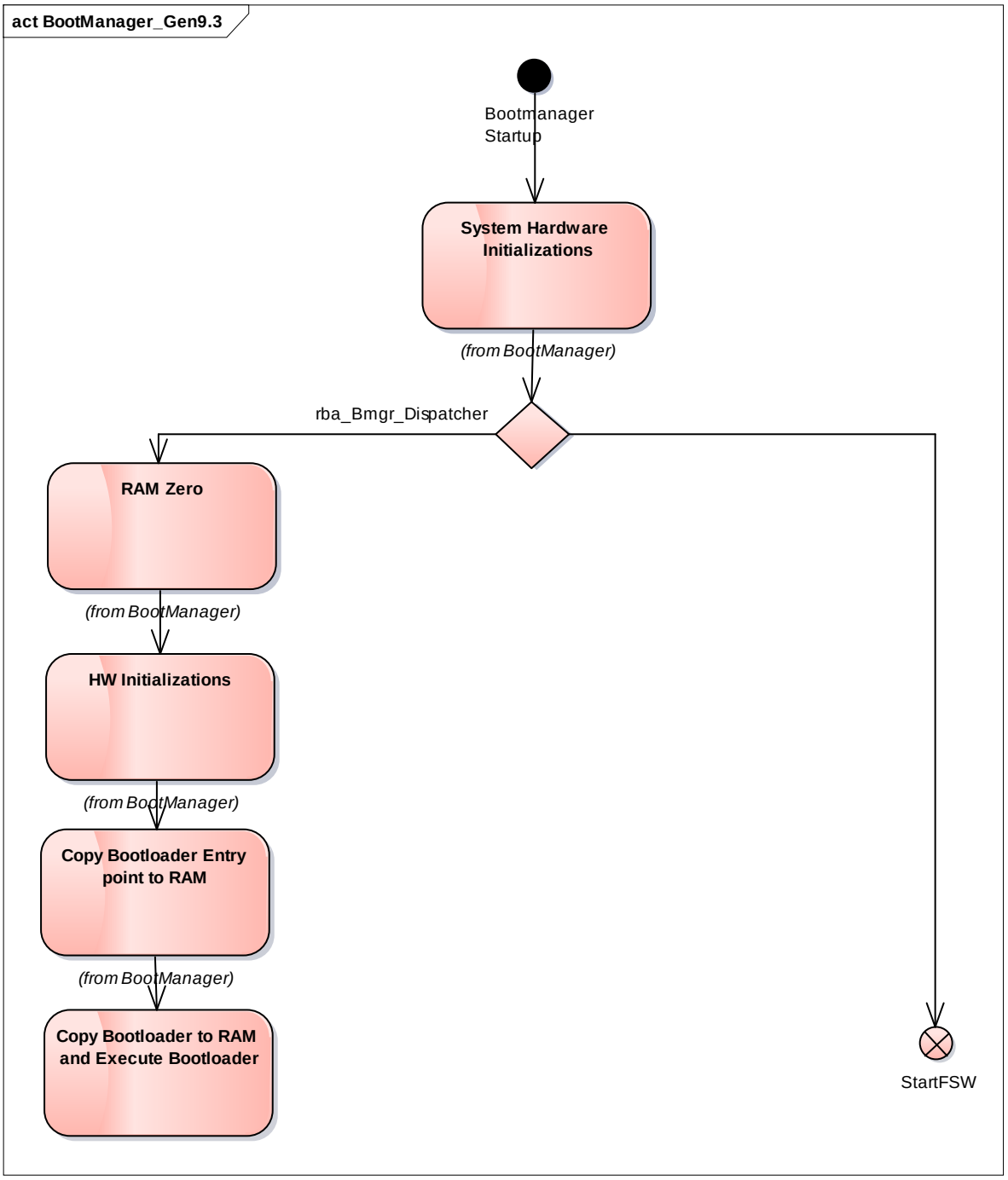
3.1.1 Bootloader Architecture



3.1.2 System Start-up

After System Startup (hard / soft reset) ECU will do some system initializations, before BootManager is called. BootManager will determine if application (FSW) is valid and in case does a direct jump to the application init functions. If not the BootManager will reset the RAM, do basic SW initializations and starts BootLoader (Boot-OS)
The following figure shows how bootmanager performs the start-up

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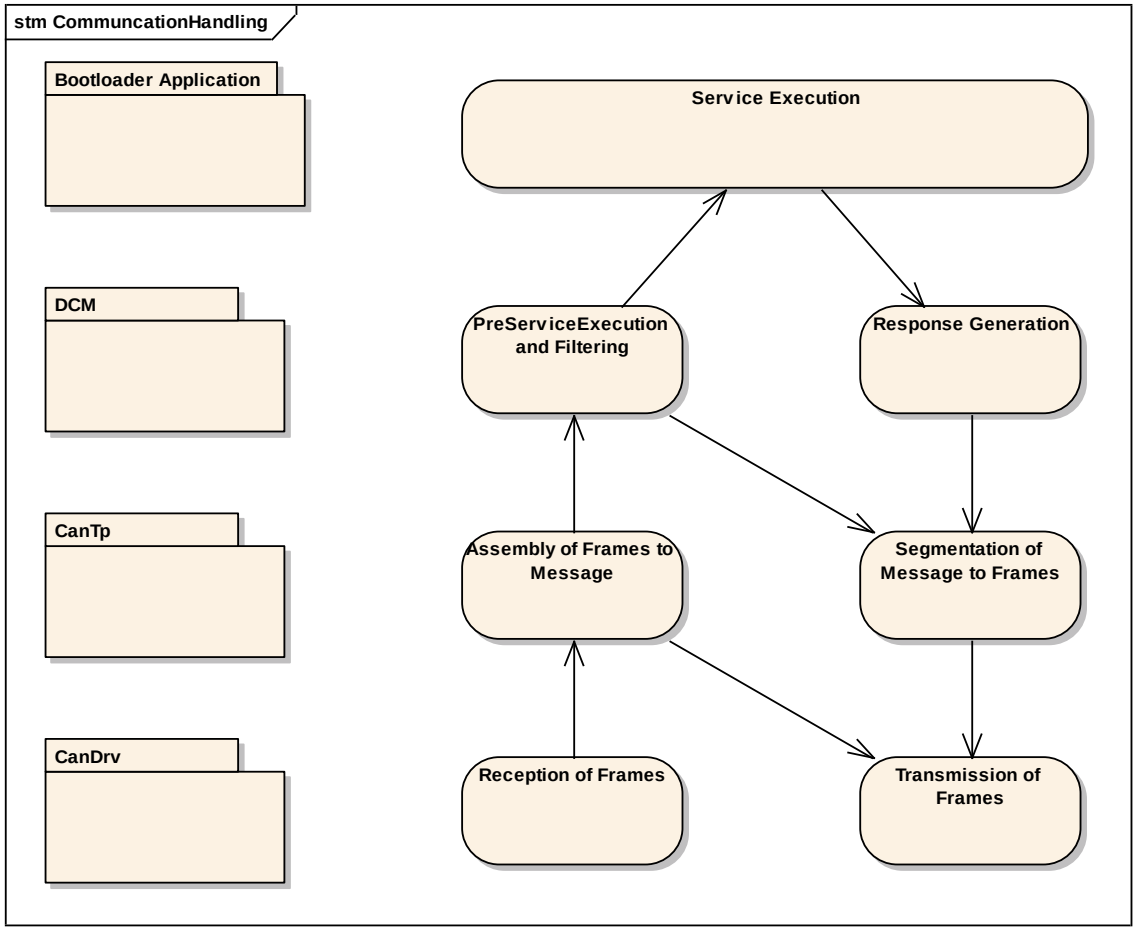


3.1.3 Diagnostic Communication

The ECU is supporting Half-Duplex communication only. This means that Communication between a diagnostic tool and the ECU is only possible in one direction at the same time.

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There is only one diagnostic protocol instance available in the ECU, which can handle only one request at a time. The rule is that any received message occupies this resource until the request message is processed.
General Communication handling (Can):



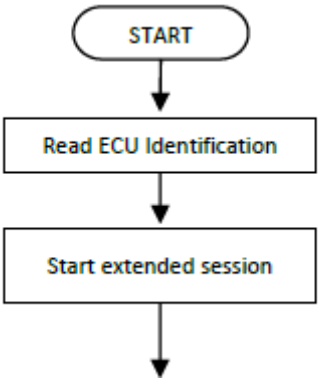
4 Software Update

The reprogramming sequence is based on the diagnostic services and divided in three programming phases.
A = Pre-programming Phase
B = Major-programming Phase
C = Post-programming Phase

4.1.1.1 Pre-Programming Steps

To download SW to a specific ECU the Tester has to perform the following steps, before a SW download:

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(a*) Read ECU Identification $22 $xx*
The ReadDataByIdentifier service is optional and used to acquire identification information from the ECU that is be reprogrammed. ECU identification information is applied by the flash process infrastructure to determine the suitable software update that shall be programmed and to document the download event.

(b*) DiagnosticSessionControl $10 $03*
The ECU shall switch to extended Mode.

4.1.1.2 Server Programming steps:

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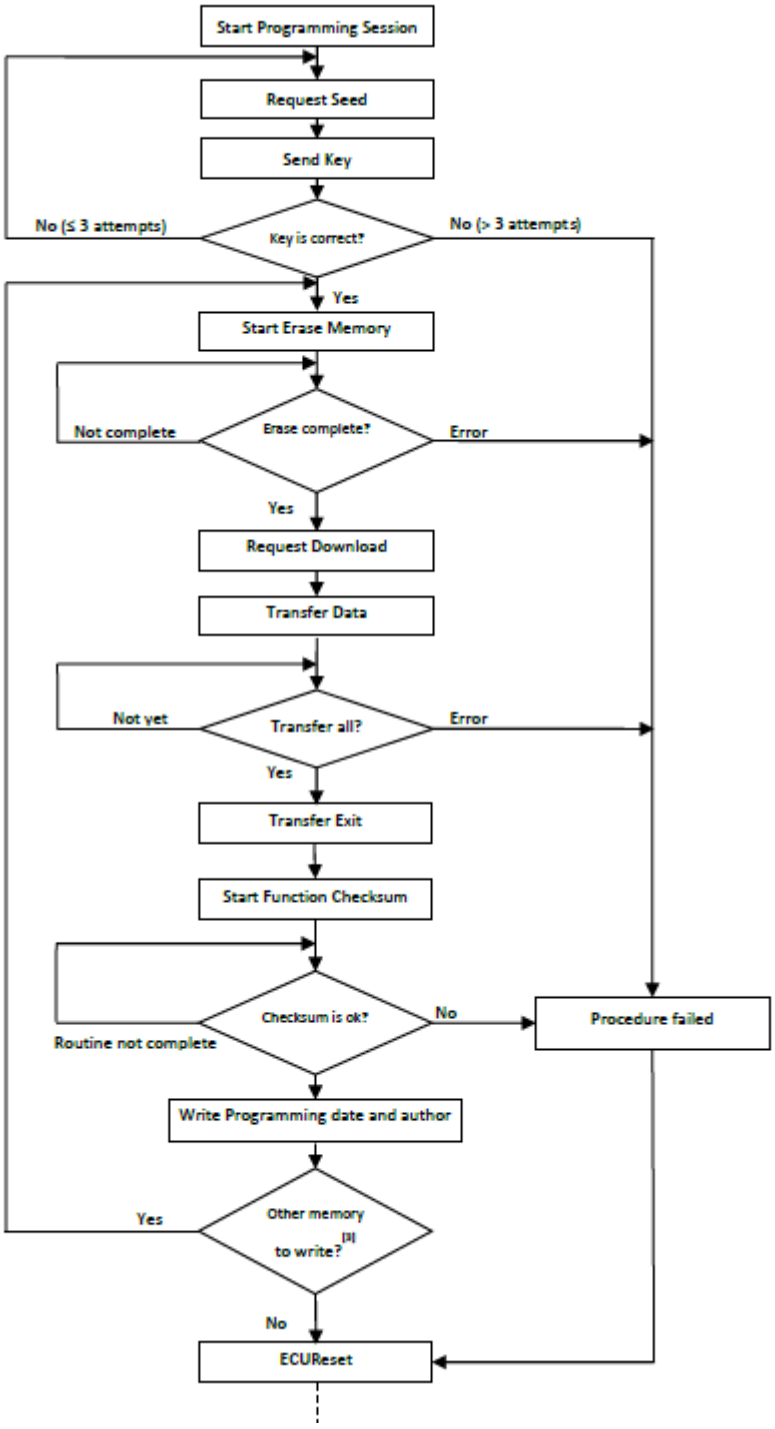


Fig. Programming step

Diagnostic Session Control $10 $02:

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The programming event is started in the ECU via a physically addressed request of the Diagnostic Session Control (10h) service with session Type equal to programming Session. When the ECU receive the request, it shall allocate all necessary resources required for programming. The positive response should be given before ECU performs the action of switching into programming session.

Security Access $27 $01/$02:
A programming event should be secured. The security access process is mandatory before download, the security access can ensure only valid tester can download the ECU. To enable Reprogramming the tester has to authenticate itself and unlock the ECU.

Routine control – Start Erase memory $31 $01 $FF $00:
The memory of the ECU shall be erased in order to allow application software or data download. This is achieved via the routine, using the Routine Control (31h) service to execute the erase routine.

***Download Application Software* 34h, 36h, 37h**: Each download of a contiguous block of application software or data to a non-volatile ECU memory location shall always follow the general data transfer method using the following service sequence:

1. RequestDownload (34h);
2. TransferData (36h);
3. RequestTransferExit (37h).

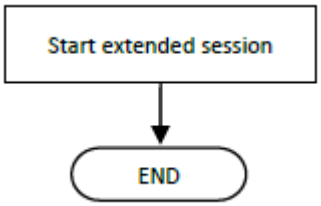
RoutineControl –"Check Memory" $31 $01 $02 $02: A RoutineControl (31h) is used to check whether the download of the logic block was successful, this function of the routine is check the integrity of download.

***Check Programming Dependencies - $31 $01 $FF $01*:** Once all application software or data blocks/modules are completely downloaded, the tester shall verify if the download has been performed successfully by initiating a routine.

Write flash reprogramming information:
This identification can be able to track a successful download. This Identifier is used to write the programming date ($2E $F1 $99) and Fingerprint information ($2E $F1 $84) to the ECU after successful download.

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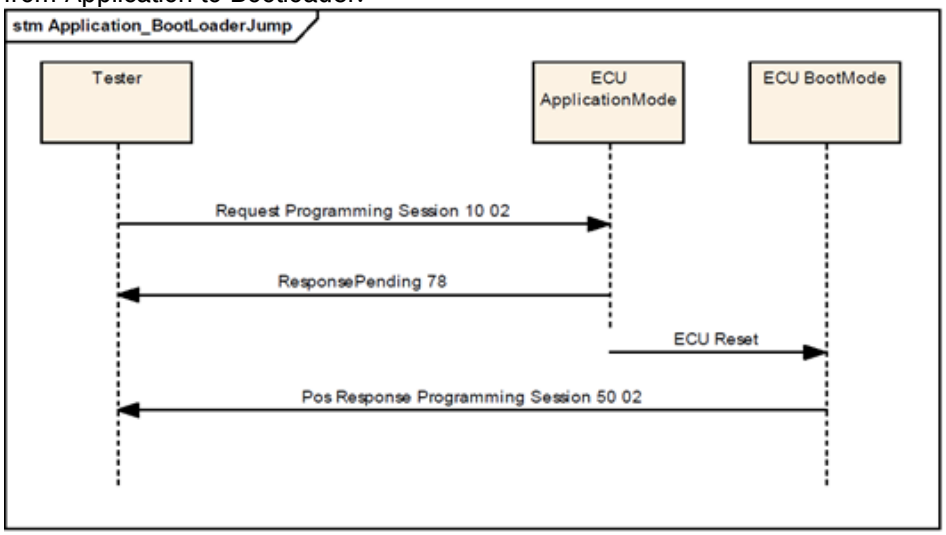
Post-Programming Steps



Switch to Extended Session:
The ECU shall switch to extended Mode.

4.1.1.3 Application to boot jump

The jump from Application to Bootloader and back from Bootloader to Application is done via an ECU reset. Bootloader and Application shall (re) init the Communication State in the correct Session. Jump from Application to Bootloader:

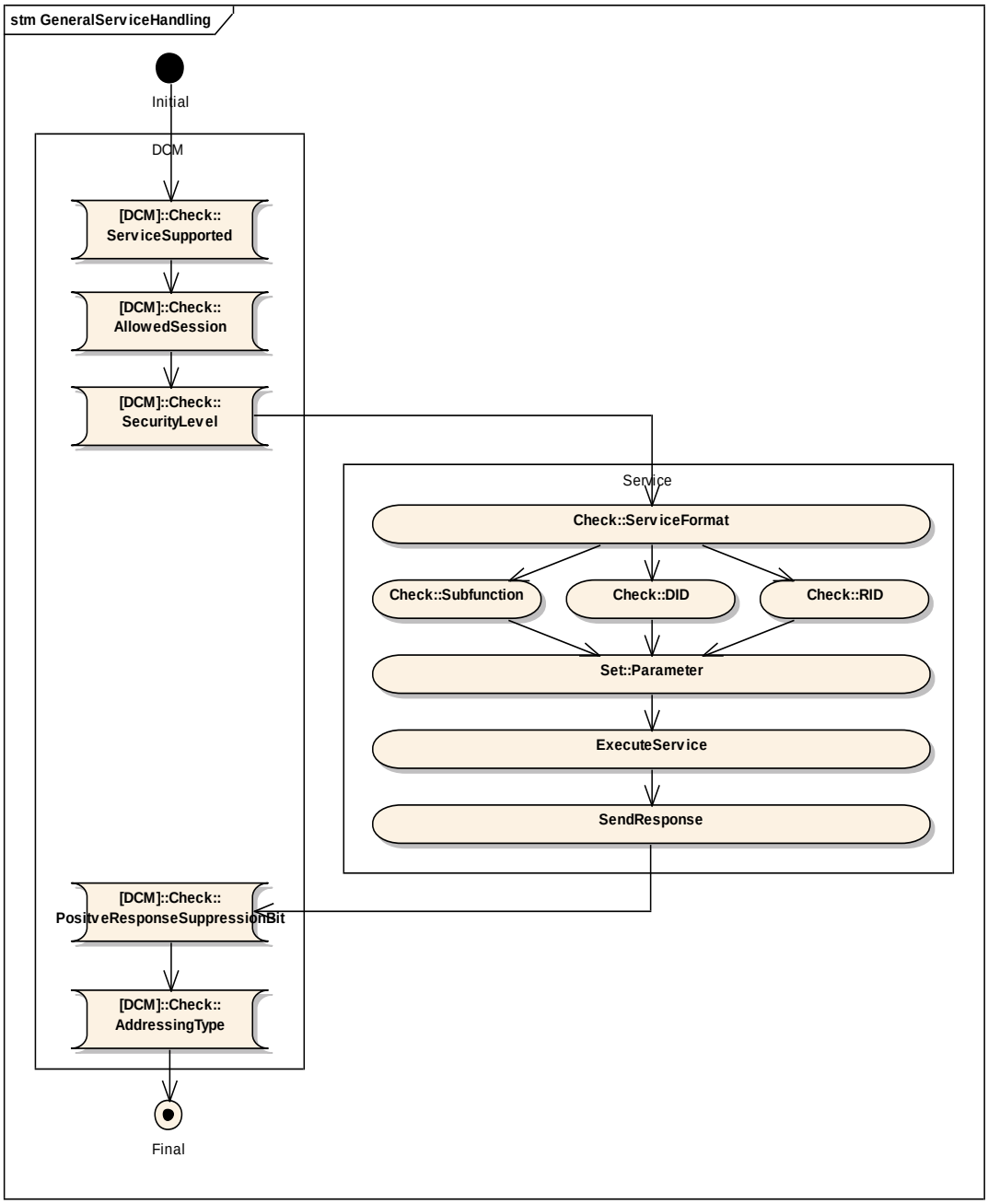


4.4 Diagnostic Behaviour

4.1.2 General Diagnostic Behaviour

General Service Execution

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General Diagnostic behaviour is implemented according to ISO14429 UDS on CAN.
Check before and after service execution are handled in DCM.
For requirements and specification please refer to AUTOSAR DCM spec.

4.1.2.1 Negative Response Codes (NRC)

–

The following table specifies all allowed negative response codes:

NRC (Hex)	Negative Response	Condition

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0x10	GeneralReject	The requested action has been rejected by the ECU. This response code shall only be implemented if none of the negative response code defined below meet the needs of the implementation.
0x11	ServiceNotSupport	Reception of a non-wanted ServiceIdentifier
0x12	SubFunctionNotSupported	The sub-function of the service is not supported
0x13	IncorrectMessageLength	Message length is invalid
0x22	ConditionsNotCorrect	Criteria of the Request are not met.
0x24	RequestSequenceError	The Service order does not correspond with the default
0x31	RequestOutOfRange	The request contains invalid Data
0x33	SecurityAccessDenied	For the service, the ECU must be unlocked
0x35	InvalidKey	The key sent does not match the server's internally calculated key.
0x36	ExceedNumberOfAttempts	The requested action will not be taken because the tester has unsuccessfully attempted to gain security access more times than the ECU security strategy allows
0x37	RequiredTimeDelayNotExpired	Delay Timer is active and a request is transmitted
0x71	TransferDataSuspended	A data transfer operation has been halted due to some fault
0x72	GeneralProgrammingFailure	This response code indicates that the server detected an error while erasing or programming a memory location in the permanent memory device (eg. FlashMemory)
0x73	WrongBlockSequenceCounter	An error has been detected in the BlockSequenceCounter Value.
0x7E	SubFunctionNotSupported - InActiveSession	The sub-function of the request is not supported in the actual session.
0x7F	ServiceNotSupported - InActiveSession	The request is not supported in the actual session.

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5 Memory layout

SI.NO	Start Address	Length	Logical Block Type
1	0x000000	0x004000	Boot Manager
2	0x004000	0x024000	Bootloader
3	0x28000	0xD8000	Application

5.1.1.1 Programming Counter

The ECU shall count the number of Erase Cycles.
The maximum number according to Programming Specification is 255 cycles.

5.1.1.2 Bootloader ROM Data table

The Bootloader Rom Data table or Bootblack Block Interface is used as a data interface which can be accessed by the application to exchange common data.

These are:

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6 Diagnostic Services supported from boot

6.1 Supported Diagnostic Services

The UDS services / functions supported by the ECU, together with the request code from tester and the corresponding response code from the ECU are listed below.
Supported services according

Diagnostic Service Description	Hex value	Phys. Req.	Funct. req	Supported in session		
Sub function Parameter				$01	$02	$03
DiagnosticSessionControl	10	x	x	x	x	x
ECUReset	11	x			x	x
ReadDataByIdentifier	22	x		x	x	x
CommunicationControl	28	x	x			x
ControlDTCSetting	85	x	x			x
WriteDataByIdentifier	2E	x		x	x	x
SecurityAccess	27	x			x	x
RequestDownload	34	x			x	
TransferData	36	x			x	
RequestTransferExit	37	x			x	
RoutineControl	31					
Start Erase Memory	31 01 FF 00	x	-	-	x	-
Start Check Routine	31 01 02 02	x	-	-	x	-
CheckProgrammingDependencies	31 01 FF 01	x	-	-	x	-

6.2 Service $10 - Diagnostic Session Control

6.2.1 Description

The "Diagnostic Session" service will switch the ECU to the requested diagnostic session.
Bootloader only differentiate between two sessions.
The bootloader default session, addressed by Diagnostic Session Type 01h
And the bootloader non-default session, addressed by Diagnostic Session Types 02h and 03h.

See also sub chapter Diagnostic Session Handling below.

The controller is passive in each diagnostic session supported by the bootloader.

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6.2.2 Availability

Supported Diagnostic Sessions	Physical request	Functional request	Supported sessions		
			$01	$02	$03
Default Session	X	X	X	X	X
programming Session	X	X	-	X	X
Extended Session	X	X	X	X	X

6.2.3 Request Message

Byte	Description	Hex Val
1	Request Service Id = [Diagnostic Session Control]	10
2	Sub Function Diagnostic Session Type [01h] = Default Diagnostic Session [02h] = Reprogramming Session [03h] = Extended Diagnostic Session Suppress Positive Response Message Indication Bit [00h] = ECU is sending a positive response message [80h] = ECU is not sending a positive response message	01, 02, 03

6.2.4 Negative Response Message

Byte	Description	Coding	Hex Val
1	Negative Response Identifier		7F
2	Request Service Id = [Diagnostic Session Control]		10
3	Negative Response Codes 12h Sub Function Not Supported This response code indicates that the requested action will not be taken because the ECU does not support the requested sub function. 13h Incorrect Message Length or Invalid Format This response code indicates that the requested action will not be taken because the length of the received request message is incorrect 22h Conditions not correct: This response code is sent when the condition to perform the diagnostic session control request is not met. 7Eh Sub-function not supported in Active session	HEX	00-FF

Note:

- The NRC 7E is especially used when $10 service is not supported in the particular session.

Session Type 01h - Default Diagnostic Session

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Byte	Description	Hex Val
1	Positive Response Service Id = [Diagnostic Session Control]	50
2	Sub Function = [Default Diagnostic Session]	01
	P2_CAN_ECU_max	
3	Byte #1 P2_CAN_ECU_max (High Byte)	00-FF
4	Byte #2 P2_CAN_ECU_max (Low Byte)	00-FF
	Format: 1 Bit = 1 ms (unsigned Hex)	
	P2*_CAN_ECU_max	
5	Byte #1 P2*_CAN_ECU_max (High Byte)	00-FF
6	Byte #2 P2*_CAN_ECU_max (Low Byte)	00-FF
	Format: 1 Bit = 10 ms (unsigned Hex)	

At ignition ON, the ECU enables the Normal diagnostic session by default.

6.2.5 Session Type 02h - Reprogramming Session

Byte	Description	Hex Val
1	Positive Response Service Id = [Diagnostic Session Control]	50
2	Sub Function = [Reprogramming Session]	02
	P2_CAN_ECU_max	
3	Byte #1 P2_CAN_ECU_max (High Byte)	00-FF
4	Byte #2 P2_CAN_ECU_max (Low Byte)	00-FF
	Format: 1 Bit = 1 ms (unsigned Hex)	
	P2*_CAN_ECU_max	
5	Byte #1 P2*_CAN_ECU_max (High Byte)	00-FF
6	Byte #2 P2*_CAN_ECU_max (Low Byte)	00-FF
	Format: 1 Bit = 10 ms (unsigned Hex)	

The ECU Flash Reprogramming Session supports the services, which are necessary to erase flash memory, download data, program and verify this data.
There is no check of preconditions (e.g. wheel speed, engine speed) available in bootloader to enter the Flash Reprogramming Session.

6.2.6 Session Type 03h - Extended Diagnostic Session

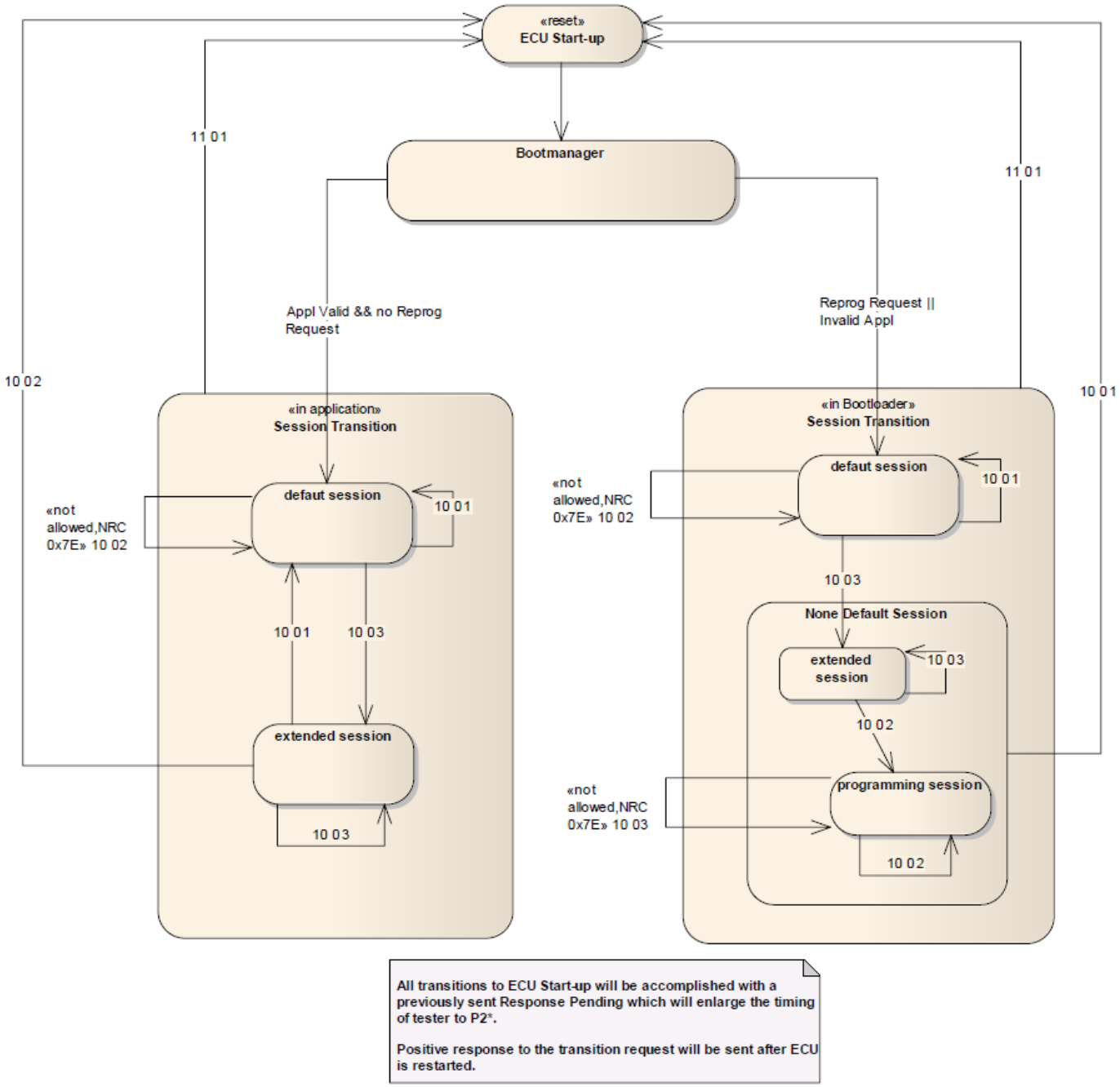
Byte	Description	Hex Val
1	Positive Response Service Id = [Diagnostic Session Control]	50
2	Sub Function = [Extended Diagnostic Session]	03
	P2_CAN_ECU_max	
3	Byte #1 P2_CAN_ECU_max (High Byte)	00-FF
4	Byte #2 P2_CAN_ECU_max (Low Byte)	00-FF
	Format: 1 Bit = 1 ms (unsigned Hex)	
	P2*_CAN_ECU_max	
5	Byte #1 P2*_CAN_ECU_max (High Byte)	00-FF
6	Byte #2 P2*_CAN_ECU_max (Low Byte)	00-FF
	Format: 1 Bit = 10 ms (unsigned Hex)	

Handled in the same way as Reprogramming Session in bootloader. Only a dummy function in bootloader, will be acknowledged positive but no further action will be taken. Only to satisfy tester sequence.

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6.2.7 Diagnostic Session Handling

The following figure depicts the required session handling of the ECU:



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6.3 Service $11 - ECUReset

6.3.1 Description

This service requests the ECU to effectively perform a reset based on the content of the reset Type parameter value in the ECU Reset request message. The ECU Reset positive response message (if required) will be sent before the reset is executed in the ECU. After a successful reset the ECU is starting the default diagnostic session.

6.3.2 Availibility

Supported Diagnostic Sessions		Default Session
	X	Reprogramming Session
	X	Extended Session
Supported Addressing Modes	X	Physical
		Functional

6.3.3 Request Message

Byte	Description	Hex Val
1	Request Service Id = [ECU Reset]	11
2	Sub Function Reset Type [01h] = Hard Reset Suppress Positive Response Message Indication Bit [00h] = ECU is sending a positive response message [80h] = ECU is not sending a positive response message	01-03

6.3.4 Negative Response Message

Byte	Description	Coding	Hex Val
1	Negative Response Identifier		7F
2	Request Service Id		11
3	Negative Response Codes 12h Sub Function Not Supported This response code indicates that the requested action will not be taken because the ECU does not support the requested sub function. 13h Incorrect Message Length or Invalid Format This response code indicates that the requested action will not be taken because the length of the received request message is incorrect 22h Conditions Not Correct This response code indicates that the requested action will not be taken, because the ECU prerequisites are not met. In this special case only if an internal error occurs.	HEX	00-FF

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6.3.5 Positive Response Message

Byte	Bit	Description	Hex Val
1		Positive Response Service Id = [ECU Reset]	51
2		Sub Function = [Reset Type] = [Hard Reset]	01-03

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6.4 Service $22 - ReadDataByIdentifier

6.4.1 Description

This service allows the diagnostic tool to request data record values from the ECU identified by Record Data Identifiers. The client request message contains one 2-byte Data Identifier values that identifies the requested Data Record.

6.4.2 Availability

Supported Diagnostic Sessions	X	Default Session
	X	Reprogramming Session
	X	Extended Session
Supported Addressing Modes	X	Physical
		Functional
Security Access		Required

Requirement:

- All EEPROM entries are protected by a checksum. If EEPROM based data cannot be retrieved from the EEPROM, the EEPROM based data for this service will be filled up with 00h (Hex values) or 30h (ASCII values).

6.4.3 Request Message

Byte	Bit	Description	Coding	Hex Val
1		Request Service Id = [Read Data By Identifier]		22
		Data Identifier	HEX	
2		Byte #1 Data Identifier High Byte		00-FF
3		Byte #2 Data Identifier Low Byte		00-FF

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6.4.4 Negative Response Message

Byte	Bit	Description	Coding	Hex Val
1	1	Negative Response Identifier		7F
2	2	Request Service Id = [Read Data By Identifier]		22
3	3	Negative Response Codes 13h Incorrect Message Length or Invalid Format This response code indicates that the requested action will not be taken because the length of the received request message is incorrect 31h Request Out Of Range This response code indicates that the requested action will not be taken because the request has incorrect parameters (e.g. Data Identifier) 22h Conditions Not Correct This response code indicates that the requested action will not be taken, because the ECU prerequisites are not met. In this special case only if an internal error occurs. 78h Request Correctly Received – Response Pending This response code indicates, that the request message was received correctly and that all parameters in the request message are valid, but the request cannot be completed within P2_CAN_ECU_max. (e.g. Access of EEPROM data)	HEX	00-FF

6.4.5 Positive Response Code

.

Byte	Bit	Description	Coding	Hex Val
1		Positive Response Service Id = [Read Data By Identifier]		62
		Data Identifier = [Programming Attempt Counter]		
2		Byte #1 Data Identifier High Byte		00-FF
3		Byte #2 Data Identifier Low Byte		00-FF
		data #1 Number of Reprogramming Attempts (High Byte)		
4		....	Hex	00-FF
5		....		
..		....	Hex	00-FF
(k-1)+4		data #k Number of Reprogramming Attempts (Low Byte)		
		K is the length defined by DID		

6.5 Service $27 - Security Access

6.5.1 Description

The service “Security Access” is used to “unlock” the ECU and gain security access. The service allows access to services requiring security protection.
The ECU is supporting a 4 Byte Hex Seed Value and a 4 Byte Key Calculation Algorithm. The Specification of the “Key Calculation Algorithm” is strictly confidential and not part of this document. Also the ECU supports two level of security access.

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6.5.2 Availability

Supported Diagnostic Sessions		Default Session
	X	Reprogramming Session
	X	Extended Session
Supported Addressing Modes	X	Physical
		Functional
Security Access		Required

Requirements:

- At ignition ON the ECU will be in “locked” security state
- Every diagnostic session transition enforces the ECU to “lock” the ECU’s security feature.
- Every “Security Access - Send Key” request must have a proceeding “Security Access - Request Seed” request.
- The ECU shall not answer to requests “Security Access - Request Seed” with a NRC “Request Sequence Error”.
- If the ECU is in “locked state” the Seed Bytes send to the diagnostic tool are in the range of 0001h ... FFFEh. If the ECU is already in “unlocked”, state all Seed Bytes will be set to zero.
- The ECU checks the received key bytes of a “Security Access - Send Key” request with the key in the ECU’s memory. If the received key bytes are valid the ECU will be unlocked. A diagnostic tool is then allowed to request services requiring security protection.
- After one unsuccessful attempt to gain Security Access the ECU memorize an unsuccessful attempt by setting a flag in non-volatile memory. After three unsuccessful attempts to gain Security Access the ECU is inserting a 10 second delay before allowing further attempts.
- In this case the unsuccessful attempt flag the 10s time delay will be also inserted after Power On / Reset.
- The 10s time delay will also be inserted if the ECU cannot determine the number of unsuccessful attempts.
- The ECU supports two Security Access Level. Each level can be unlocked individually.
- A successful attempt clears the flag for memorizing unsuccessful attempts.

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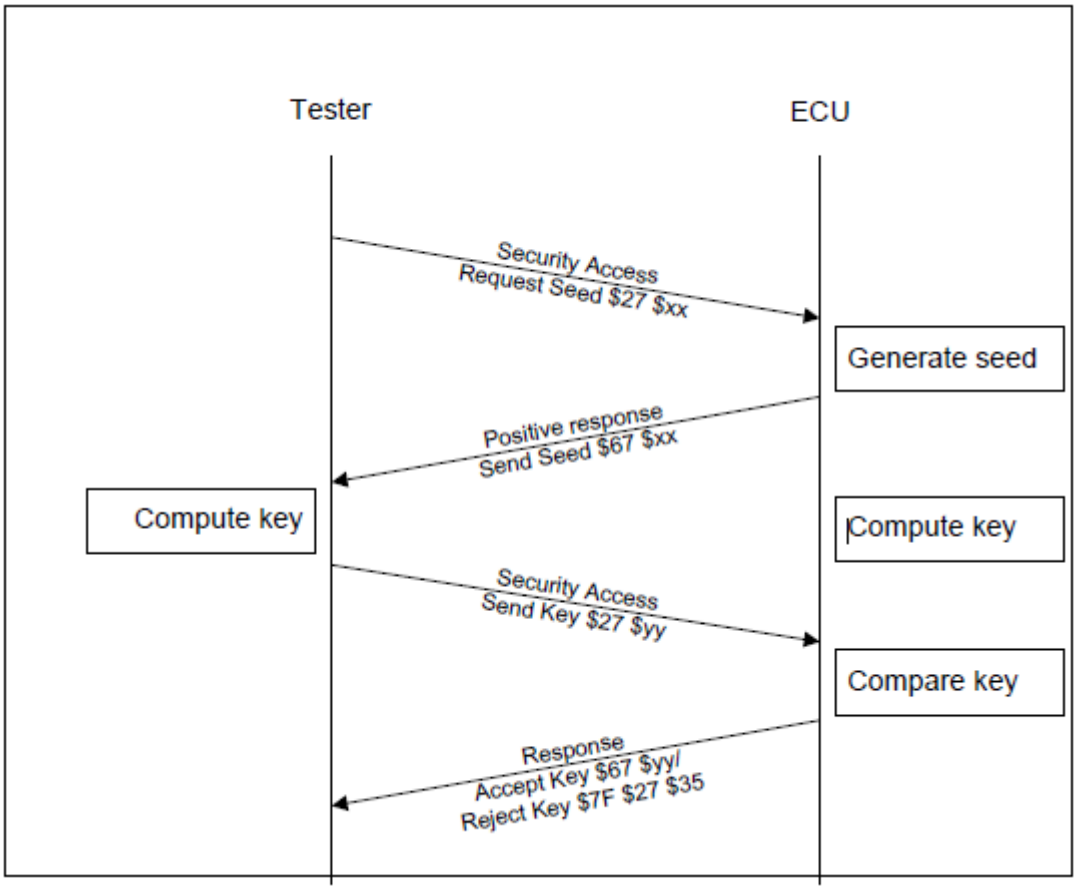


Fig. Security access procedure

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6.5.3 Negative Response Message

Byte	Description	Coding	Hex Val
1	Negative Response Identifier		7F
2	Request Service Id = [Security Accessl]		27
3	Negative Response Codes 12h **Sub Function Not Supported** This response code indicates that the requested action will not be taken because the ECU does not support the requested sub function 13h **Incorrect Message Length or Invalid Format** This response code indicates that the requested action will not be taken because the length of the received request message is incorrect 22h **Conditions Not Correct** This response code indicates that the requested action will not be taken, because one of the following conditions are not correct (see session specific session enable conditions) or a time-out has appeared. 24h **Request Sequence Error** This response code indicates that a "Send Key" is requested without a corresponding "Request Seed". 35h **Invalid Key** This response code indicates that the ECU has not given access because the key sent by the diagnostic tool did not match with the key in the ECU's memory 36h **Exceed umber of attempts** This response code is sent when the delay timer is activated because of failure to access after exceeding the maximum allowable number of attempts. 37h **Delay requirements are not satisfied** This response code is sent when the request message of this service is received as the delay timer is activated.	HEX	00-FF

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6.5.4 Security Access Type - Request Seed

Request Message - Request Seed

Byte	Bit	Description	Hex Val
1		Request Service Id = [Security Access]	27
2	0-7	Sub Function Security Access Type = Request Seed [01] = Extended security Level	01

Positive Response Message - Request Seed

Byte	Bit	Description	Coding	Hex Val
1		Positive Response Service Id = [Security Access]		67
2		Sub Function = [Security Access Type = Request Seed] Sub Function [01] = Extended security Level		01
		Security Seed		
3		Byte #1 Security Seed (High Byte)	HEX	00-FF
4		Byte #2 Security Seed	HEX	00-FF
5		Byte #3 Security Seed	HEX	00-FF
6		Byte #4 Security Seed (Low Byte)	HEX	00-FF
		Range: [0000h] = Security Seed in case ECU is unlocked [0001h] …[FFFEh] = Security Seed in case ECU is locked [FFFFh] = Not Defined		

6.5.5 Security Access Type - Send Key

Request Message - Send Key

Byte	Bit	Description	Coding	Hex Val
1		Request Service Id = [Security Access]		27
2	0-7	Sub Function Security Access Type = Send Key [02] = Extended security Level		02
		Security Key		
3		Byte #1 Security Key (High Byte)	HEX	00-FF
4		Byte #2 Security Key	HEX	00-FF
5		Byte #3 Security Key	HEX	00-FF
64		Byte #4 Security Key (Low Byte)	HEX	00-FF

Positive Response Message - Send Key

Byte	Bit	Description	Coding	Hex Val
1		Positive Response Service Id = [Security Access]		67
2	0-7	Sub Function Security Access Type = Send Key [02] = Extended security Level		02

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6.6 Service $28 - Communication Control

6.6.1 Description

The purpose of this service is to switch on/ off the transmission of ECU messages (e.g. application communication messages)
In bootloader operation this service is not supported because the bootloader itself doesn't send periodical messages.

6.6.2 Availability

Availability acc. Flash Bootloader_V1.4

Supported Diagnostic Sessions		Default Session
		Reprogramming Session
	x	Extended Session
Supported Addressing Modes	x	Physical
	x	Functional
Security Access		Required

6.6.3 Request Message

Byte	Description	Coding	Hex Val
1	Request Service Id = [Communication Control]		28
2	Sub Function Control Type [xxxxxxx0b] = EnableRxAndEnableTx [xxxxxx11b] = DisableRxAndDisableTx Suppress Positive Response Message Indication Bit [0xxxxxxxb] = ECU is sending a positive response message [1xxxxxxxb] = ECU is not sending a positive response message		00-FF
3	Communication Type Normal Communication Messages [01h]	HEX	00-FF

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6.6.4 Negative Response Message

Byte	Bit	Description	Coding	Hex Val
1		Negative Response Identifier		7F
2		Request Service Id = [Communication Control]		28
3		Negative Response Codes 12h **Sub Function Not Supported** This response code indicates that the requested action will not be taken because the ECU does not support the requested sub function 13h **Incorrect Message Length or Invalid Format** This response code indicates that the requested action will not be taken because the length of the received request message is incorrect 31h **Request Out Of Range** This response code indicates that the requested action will not be taken because the request has incorrect parameters (e.g. Wrong communication Type parameter)	HEX	00-FF

6.6.5 Positive Response Message

Byte	Bit	Description	Coding	Hex Val
1		Positive Response Service Id = [Communication Control]		68
2		Sub Function = [Control Type]		00-7F

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6.7 Service $2E - Write Data by Identifier

6.7.1 Description

This service allows the diagnostic tool to write data records into the ECU identified by Record Data Identifiers. The client request message contains one 2-byte Data Identifier values that identifies the requested Data Record.

6.7.2 Availability

Supported Diagnostic Sessions	x	Default Session
	x	Reprogramming Session
	x	Extended Session
Supported Addressing Modes	x	Physical
		Functional

Requirements:

- All EEPROM based data are secured with a Checksum

6.7.3 Request Message

Byte	Bit	Description	Coding	Hex Val
1		Request Service Id = [WriteDataByIdentifier]		2E
		Data Identifier	HEX	
2		Byte #1 Data Identifier High Byte		00-FF
3		Byte #2 Data Identifier Low Byte		00-FF
4		Data Record	HEX	00-FF
…		Data #1		
(k-1)+4		….		00-FF
		Data #k		
		K is the length defined by DID		

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Supported Data Identifiers:

<table>
<tr><th>Data Identifier</th><th>Description</th></tr>
<tr><td>0xF199</td><td>Programming date
<table>
<tr><td colspan="3">Tester ⇒ ECU: WriteDataByLocalIdentifierRequest</td></tr>
<tr><td>Byte</td><td>Content</td><td>Value</td></tr>
<tr><td>1</td><td>WriteDataByIdentifierSID</td><td>2EH</td></tr>
<tr><td>2</td><td>Data Identifier High Byte</td><td>F1H</td></tr>
<tr><td>3</td><td>Data Identifier Low Byte</td><td>99H</td></tr>
<tr><td>4</td><td>Programming Date Year (YY) MSB</td><td>XX BCD</td></tr>
<tr><td>5</td><td>Programming Date Month (MM)</td><td>XX BCD</td></tr>
<tr><td>6</td><td>Programming Date Day (DD) LSB</td><td>XX BCD</td></tr>
</table></td></tr>
<tr><td>0xF184</td><td>Write Fingerprint
<table>
<tr><td colspan="3">Tester ⇒ ECU: WriteDataByLocalIdentifierRequest</td></tr>
<tr><td>Byte</td><td>Content</td><td>Value</td></tr>
<tr><td>1</td><td>WriteDataByIdentifierSID</td><td>2EH</td></tr>
<tr><td>2</td><td>Data Identifier High Byte</td><td>F1H</td></tr>
<tr><td>3</td><td>Data Identifier Low Byte</td><td>84H</td></tr>
<tr><td>4</td><td>Reprogram Part Number Position #1 (MSB)</td><td>ASCII</td></tr>
<tr><td>5</td><td>Reprogram Part Number Position # 2</td><td>ASCII</td></tr>
<tr><td>6</td><td>Reprogram Part Number Position # 3</td><td>ASCII</td></tr>
<tr><td>7</td><td>Reprogram Part Number Position # 4</td><td>ASCII</td></tr>
<tr><td>8</td><td>Reprogram Part Number Position # 5</td><td>ASCII</td></tr>
<tr><td>9</td><td>Reprogram Part Number Position # 6</td><td>ASCII</td></tr>
<tr><td>10</td><td>Reprogram Part Number Position # 7</td><td>ASCII</td></tr>
<tr><td>11</td><td>Reprogram Part Number Position # 8</td><td>ASCII</td></tr>
<tr><td>12</td><td>Reprogram Part Number Position # 9</td><td>ASCII</td></tr>
<tr><td>13</td><td>Reprogram Part Number Position #10</td><td>ASCII</td></tr>
<tr><td>14</td><td>Reprogram Part Number Position #11</td><td>ASCII</td></tr>
<tr><td>15</td><td>Reprogram Part Number Position #12 (LSB)</td><td>ASCII</td></tr>
</table></td></tr>
<tr><td>0xF194</td><td>BBNumber
<table>
<tr><td colspan="4">Tester ⇒ ECU: WriteDataByLocalIdentifierRequest</td></tr>
<tr><td>Byte</td><td colspan="2">Contents</td><td>Value</td></tr>
<tr><td>1</td><td colspan="2">WriteDataByIdentifierSID</td><td>2EH</td></tr>
<tr><td>2</td><td colspan="2">Data Identifier High Byte</td><td>F1 H</td></tr>
<tr><td>3</td><td colspan="2">Data Identifier Low Byte</td><td>94 H</td></tr>
<tr><td>4</td><td rowspan="3">SW BB-Number (ASCII)</td><td>Figure 1</td><td>xx H</td></tr>
<tr><td>5</td><td>Figure 2</td><td>xx H</td></tr>
<tr><td>6</td><td>Figure 3</td><td>xx H</td></tr>
</table></td></tr>
</table>

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6.7.4 Positive Response Message

Byte	Bit	Description	Coding	Hex Val
1		Positive Response Service Id = [Write Data By Identifier]		6E
		Data Identifier		
2		Byte #1 Data Identifier High Byte	Hex	00-FF
3		Byte #2 Data Identifier Low Byte	Hex	00-FF

6.7.5 Negative Response Message

Byte	Bit	Description	Coding	Hex Val
1		Negative Response Identifier		7F
2		Request Service Id = [Write Data By Identifier]		2E
3		Negative Response Codes	HEX	00-FF
		13h **Incorrect Message Length or Invalid Format** This response code indicates that the requested action will not be taken because the length of the received request message is incorrect		
		22h **Conditions Not Correct** This response code indicates that the ECU has detected an error when reading or writing an EEPROM memory location.		
		31h **Request Out Of Range** This response code indicates that the requested action will not be taken because the request has incorrect parameters. (e.g. wrong Data Identifier, wrong parameter in data record)		
		33h **Security Access Denied** This response code indicates that the requested action will not be taken because the requested action is security protected and requires unlocking the ECU.		
		72h **General programming failure** This response code is sent when the ECU detects an error in writing data to the memory cell.		

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6.8 Service $31 – Routine Control

The service, “Routine Control”, is used by the tester to start a routine in the memory of the ECU.

6.8.1 Description

This service allows the diagnostic tool to start routines related to the reprogramming sequence.

6.8.2 Availability

Routine control	Physical request	Functional request	Supported sessions		
			$01	$02	$03
Start Erase Memory	**X**	-	**-**	**X**	**-**
Start check routine	**X**	-	-	**X**	-
Check programming dependencies	**X**	-	-	**X**	-

6.8.3 Request Message

Byte	Bit	Description	Coding	Hex Val
0		Request Service Id = [Routine control]		31
1		Routine Control Type 01 – Start Routine	HEX	01
2 3		Routine Identifier 1. check routine 2. Check programming preconditions 3. erase memory 4. check programming dependencies	HEX	0202 FF00 FF01
4 … #n		Routine Control Option Record Routine Control Option #1 ….. Routine Control Option #m	HEX	00-FF

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6.8.4 Negative Response Codes

Byte	Bit	Description	Coding	Hex Val
1		Negative Response Identifier		7F
2		Request Service Id = [Diagnostic Session Control]		31
3		Negative Response Codes 12h **Sub Function Not Supported** This response code indicates that the requested action will not be taken because the ECU does not support the requested sub function. 13h **Incorrect Message Length or Invalid Format** This response code indicates that the requested action will not be taken because the length of the received request message is incorrect 22h **Conditions Not Correct** This response code indicates that the requested action will not be taken, because one of the following conditions are not correct (see session specific session enable conditions) or a time-out has appeared. 24h **Request Sequence Error** This response code indicates that a RoutineControl is requested out of defined reprogramming sequnce". 31h **Request Out Of Range** This response code indicates that the requested action will not be taken because the request has incorrect parameters 33h **Security Access Denied** This response code indicates that the requested action is security protected and requires unlocking the ECU.	HEX	00-FF

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6.8.5 Routine Control Identifiers supported in Bootloader

The routine control service in Bootloader supports routine identifiers:

1. $02 $02- "check routine"
2. $FF $00- “erase memory“
3. $FF $01- "check programming dependencies".

Bootloader must only support the routine control type 0x01- “startup routine”.

6.8.5.1 Identifier $02 $02 – Check Routine

Byte	Description	Hex Val
0	Request Service Id = [Diagnostic Session Control]	31
1	Request Control Type- Start routine	01
2 3	Routine Identifier [ ] CheckRoutine #1 (MSB) CheckRoutine #2 (LSB)	02 02
4 5 6 7 8 9 10	AddressLengthFormatID BlockID Length of Checksum Checksum byte 1 Checksum byte 2 Checksum byte 3 Checksum byte 4	11 01 04 xxH xxH xxH xxH

6.8.5.2 Positive Response Message

Byte	Bit	Description	Coding	Hex Val
0		Positive Response Service Identifier		71
1		Sub Function = [Routine Control Type] = [CheckRoutine]		01
2 3		Routine Identifier Byte #1 Routine Identifier High Byte Byte #2 Routine Identifier Low Byte		 02 02
4		Routine Status Record [00h] = Correct Result [01h] = Incorrect Result		00-01

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6.8.5.3 Identifier $FF $00 – Erase Memory

Byte	Description	Hex Val
0	Request Service Id = [Diagnostic Session Control]	31
1	Request Control Type	01
2 3	Routine Identifier [ ] CheckRoutine #1 (MSB) CheckRoutine #2 (LSB)	FF 00
4 5	Address and Length Format Identifier BlockId	01 01

6.8.5.4 Positive Response Message

Byte	Bit	Description	Coding	Hex Val
1		Positive Response Service Identifier		71
2		Sub Function = [Routine Control Type] = [CheckRoutine]		01
3 4		Routine Identifier eraseMemory #1 (MSB) eraseMemory #2 (LSB)		 FF 00
5		Routine Status Record [00h] = Correct Result [01h] = Incorrect Result		00-01

6.8.5.5 Identifier $FF $01 – Check programming dependencies

- This service is used for check if the compatibility dependencies are fulfilled.

Byte	Description	Hex Val
1	Request Service Id = Request Service Identifier	31
2	Request Control Type	01
3 4	Routine Identifier [ ] checkProgrammingDependencies #1 (MSB) checkProgrammingDependencies #2 (LSB)	FF 01

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6.8.5.6 Positive Response Message

Byte	Bit	Description	Coding	Hex Val
1		Positive Response Service Id = [Routine Control]		71
2		Sub Function = [Routine Control Type] = [Start Routine]		01
		Routine Identifier		
3		checkProgrammingDependencies #1 (MSB)		FF
4		checkProgrammingDependencies #2 (LSB)		01
5		Routine Status Byte		
		[00h] = Dependencies Check passed - No Failure		00
		[01h] = Dependencies Check failed – internal check routine error		01

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6.9 Service $34 – RequestDownload

6.9.1 Description

This service is used to initiate a download procedure.
ECU can only accept 512 bytes at a time in a command.

6.9.2 .Availibility

Supported Diagnostic Sessions		Default Session
	X	Reprogramming Session
		Extended Session
Supported Addressing Modes	**X**	Physical
		Functional

6.9.3 Request Message

Byte	Description	Hex Val
0	Request Service Id = [Request Download]	34
1	Data format Identifier	00
2	AddressAndLengthFormatIden tifier	41
3	Block ID	01
4 -7	Length [L1] MS byte Length [L2] Length [L3] Length [L4] LS byte	YY .. .. YY

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6.9.4 Negative Response Message

Byte	Bit	Description	Coding	Hex Val
1		Negative Response Identifier		7F
2		Request Service Id = [Diagnostic Session Control]		34
3		Negative Response Codes 13h **Incorrect Message Length or Invalid Format** This response code indicates that the requested action will not be taken because the length of the received request message is incorrect 22h **Conditions not correct** This return code shall be sent if the server received a request for this service while in the process of receiving a download of a software or calibration module. This will occur due to size mismatch. 31h **Request Out Of Range** This response code should be transmitted in case of the following conditions: 1. DataFormatIdentifier is invalid 2. AddressAndLengthFormatIdentifier is invalid 3. MemoryAdderss/memory size is invalid 33h **Security Access Denied** This response code indicates that the requested action will not be taken because the requested action is security protected and requires unlocking the ECU. 70h **Upload/download not accepted** This response code indicates, that an attempt to download to a server memory cannot be accomplished due to some of the fault conditions.		00-FF

6.9.5 Positive Response Message

Byte	Bit	Description	Coding	Hex Val
1		Positive Response Service Id		74
2		Length Format Identifier		20
3		Max Number Of Block Length byte (MSB)		02
4		byte (LSB)		00

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6.10 Service $36 – TransferData

6.10.1 Description

This service is used to transfer the data from the tester to the ECU (download).

6.10.2 Availibility

Supported Diagnostic Sessions		Default Session
	X	Reprogramming Session
		Extended Session
Supported Addressing Modes	**X**	Physical
		Functional

6.10.3 Request Message

Byte	Bit	Description	Hex Val
1		Request Service Id = [Transfer Data]	36
2		blockSequenceCounter	00-FF
3		TransferRequestParameterRecord[transferRequestParameter#1]	00-FF
:			:
n		TransferRequestParameterRecord[transferRequestParameter#n	00-FF

6.10.4 Positive Response Message

Byte	Bit	Description	Hex Val
1		Positive Response Service Id = [TransferData]	76
2		BlockSequenceCounter	00-FF

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6.10.5 Negative Response Message

Byte	Bit	Description	Coding	Hex Val
1		Negative Response Identifier		7F
2		Request Service Id = [Diagnostic Session Control]		36
3		Negative Response Codes 13h **Incorrect Message Length or Invalid Format** This response code indicates that the requested action will not be taken because the length of the received request message is incorrect 24h **Request Sequence Error** This response code is sent in the following conditions: 1. The request for download or upload service is not valid when the service request is received. 2. If the request for download or upload service is valid, but the ECU has received all the data determined by the memory size parameters in the request for download or upload service. 31h **Request Out Of Range** If the transferRequestParameterRecord contains additional control parameters and the control information is not available, the response code should be sent. 71h **Data transmission suspended** This response code should be sent in the following conditions: 1. The response code indicates that the operation of data transmission is suspended due to some error. 2. The length of the download module does not meet the memorySize parameter requirements in the request message of the request for download service. 72h **General programming error** This response code should be sent if the ECU detects any error while deleting or programming in the permanent memory during the download of data.. 73h **Block sequence counter error** This response code should be sent if the ECU detects any error in the blockSequenceCounter sequence. 22h **Condition not correct** This response code is sent when the condition to perform the diagnostic session control requirement is not met.	HEX	00-FF

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6.11 Service $37 - RequestTransferExit

This service is used to terminate a download.

6.11.1 Availability

Supported Diagnostic Sessions		Default Session
	X	Reprogramming Session
		Extended Session
Supported Addressing Modes	**X**	Physical
		Functional

6.11.2 Request Message

Byte	Bit	Description	Hex Val
1		Request Service Id = [Request Transfer Exit]	37

6.11.3 Positive Response Message

Byte	Bit	Description	Hex Val
1		Positive Response Service Id = [RequestTransferExit]	77

6.11.4 Negative Response Message

Byte	Bit	Description	Coding	Hex Val
1	1	Negative Response Identifier		7F
2	2	Request Service Id = [Read Data By Identifier]		37
3	3	Negative Response Codes 13h **Incorrect Message Length or Invalid Format** This response code indicates that the requested action will not be taken because the length of the received request message is incorrect 24h **Request sequence error** This response code should be transmitted in case of the following conditions: 1. The programming process has not been completed when this service request is received. 2. The request for download or upload service is not valid. 78h **Request Correctly Received – Response Pending** This response code indicates, that the request message was received correctly and that all parameters in the request message are valid, but the request cannot be completed within P2_CAN_ECU_max. (e.g. Access of EEPROM data)	HEX	00-FF

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6.12 Service $3E - Tester Present

6.12.1 Description

This service is used to indicate to an ECU that the diagnostic tool is still connected to the vehicle and that diagnostic sessions are to remain active.

6.12.2 Availability

Supported Diagnostic Sessions	X	Default Session
	X	Reprogramming Session
	X	Extended Session
Supported Addressing Modes	X	Physical
	X	Functional

Request Message

Byte	Bit	Description	Coding	Hex Val
1		Request Service Id = [Tester Present]		3E
2	0-6 7	Sub Function Function Type [00h] = Zero Sub Function Suppress Positive Response Message Indication Bit [00000000h] = ECU is sending a positive response message [10000000h] = ECU is not sending a positive response message	HEX	00/80

6.12.3 Positive Response Message

Byte	Bit	Description	Coding	Hex Val
1		Positive Response Service Id = [Tester Present]		7E
2		Sub Function = [Zero Sub Function]		00

6.12.4 Negative Response Message

Byte	Bit	Description	Coding	Hex Val
1		Negative Response Identifier		7F
2		Request Service Id = [Tester Present]		3E
3		Negative Response Codes 12h **Sub Function Not Supported** This response code indicates that the requested action will not be taken because the ECU does not support the requested sub function 13h **Incorrect Message Length or Invalid Format** This response code indicates that the requested action will not be taken because the length of the received request message is incorrect		00-FF

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6.13 Service $85 - Control DTC Setting

6.13.1 Availability

Availability acc. Flash Bootloader_V1.4

Supported Diagnostic Sessions		Default Session
		Reprogramming Session
	x	Extended Session
Supported Addressing Modes	x	Physical
	x	Functional

Requirement:

- In bootloader operation only dummy functionality to satisfy tester reprogramming sequence.

6.13.2 Request Message

Byte	Description	Hex Val
1	Request Service Id = [Control DTC Setting]	85
2	Sub Function = [DTC Setting Type]	00-7F

6.13.3 Response Message

Byte	Bit	Description	Hex Val
1		Positive Response Service Id = [Control DTC Setting]	C5
2		Sub Function = [DTC Setting Type]	00-7F

6.13.4 Negative Response Codes

	Bit	Description	Coding	Hex Val
1		Negative Response Identifier		7F
2		Request Service Id = [Control DTC Setting]		85
3		Negative Response Codes 12h **Sub Function Not Supported** This response code indicates that the requested action will not be taken because the ECU does not support the requested sub function 13h **Incorrect Message Length or Invalid Format** This response code indicates that the requested action will not be taken because the length of the received request message is incorrect	Hex	00-FF

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6.14 SuppressPositiveResponse bit

The client (e.g., tester) could set the SuppressPosResponse bit to for the following UDS services:

1. Communication Control (28hex)
2. TesterPresent (3Ehex)
3. ControlDTCSetting (85hex)

Other than this services will give NRC 12 (Sub function not supported) as a suppress response. Also, NRC 7F and 7E is not suppressed during functional.

7 Requirements to Sub-FNIDs

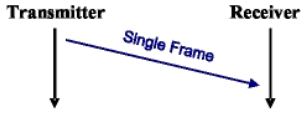
7 Transport Protocol: CAN (SubFNID 689)

7.1.1 Data Stream

The ECU supports CAN diagnostic messages using the Unacknowledged Segmented Data Transfer (USDT) mechanism as specified in ISO 15765-2. Accordingly the following two types of CAN data streams are supported:

Non-Segmented Data Stream

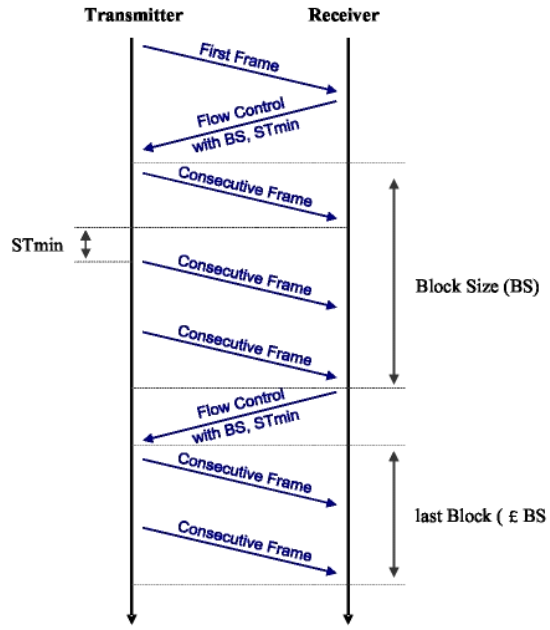
A non-segmented data stream consists of a Single Frame (SF). Non-segmented data stream are used for messages with a maximum data length of 7 data bytes in case of normal addressing



Segmented Data Stream

A segmented data stream consists of multiple frames of segmented CAN messages. A segmented data stream must be used if the maximum data length exceeds 7 data bytes in case of normal addressing.
The data stream comprises a First Frame (FF) and a number of Consecutive Frames (CF) containing the subsequent data bytes, following the First Frame. The receiver has to send a Flow Control (FC) after reception of the First Frame.
The Flow Control contains information regarding Block Size and the Separation Time. The Block Size (BS) is the maximum number of Consecutive Frames the receiver allows the sender to send, before waiting for an authorisation (Additional Flow Control Frames) from the receiver to continue transmission. The Separation Time (Stmin) is the (minimum) time the sender has to wait between transmissions of two consecutive frames.

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7.1.2 Message Types

Each Single Frame (SF), First Frame (FF), Consecutive Frame (CF) or Flow Control (FC) has an 8-byte Data Field (DLC of CAN message is set to 8) consisting of the following parts:

- Protocol Control Information (PCI)

indicates the frame type, number of frames, data length etc.

- Transmission data (Data)

contains data to be sent

Data Field Structure					
Frame Type	**Byte 1**		**Byte 2**	**Byte 3**	**Byte 4 – 8**
	Bit 7- 4	**Bit 3 – 0**			
Single Frame (SF)	Type_SF = 0000B	SF_DL	DB 1	DB 2	DB 3 – 7
First Frame (FF)	Type_FF = 0001B	FF_DL		DB 1	DB 2 – 6
Consecutive Frame (CF)	Type_CF = 0010B	SN	DB x_1	DB x_2	DB x_3-x_7
Flow Control (FC)	Type_FC = 0011B	FS	BS	Stmin	N/A

SF_DL: 4 Bit Single Frame Data length
SN: Sequence Number
BS: Block Size
DB: Data Byte

FF_DL: 12 Bit First Frame Data length
FS: Status of data stream
Stmin: Minimum Separation Time

Single Frame

A Single Frame message type is used for data messages with a maximum data length of 7 data bytes. The PCI Type 0000B identifies a Single Frame. The Single Frame data length SF_DL is defined as:

Single Frame Data Length (SF_DL):

Hex Value	Description
0h – 7h	Single Frame Data length (SF_DL) contains number of data Bytes

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First Frame

Messages longer than 7 data bytes are segmented into a First Frame and one or more Consecutive Frames. The PCI Type 0001B identifies a First Frame. The First Frame data length FF_DL is defined as:

First Frame Data Length (FF_DL):

Hex Value	Description of FF_DL
000h – 007h	Reserved
008h – FFFh	First Frame Data length (FF_DL) contains number of data bytes. 12 Bit FF_DL theoretically supports sending of up to 4096 data bytes. Note: The ECU supports segmented data messages up to maximally 50 data bytes.

Consecutive Frame

Consecutive Frames are following a First Frame. The PCI Type 0010B identifies a Consecutive Frame. The Sequence Number (SN) in the lower nibble of a Consecutive Frame is specified as follows:

- The first Consecutive Frame always starts with Sequence Number 1.
- The Sequence Number is incremented by one for each additional Consecutive Frame.
- The Sequence Number (SN) value is not affected by any Flow Control (FC) frame.
- When the Sequence Number reaches the value of 15, the next Consecutive Frame will be sent with a Sequence Number set to 0.

Definition of Sequence Number (SN):

Protocol Data name	FF	CF	CF	CF	CF	CF	CF	CF	CF
SN (hex)	--	1	2	...	E	F	0	1	...

Hex Value	Description
0h – Fh	Sequence Number (SN) The Sequence Number has a range of 0 to 15. The Sequence Number is stored in the lower 4 bits of Byte 1.

Flow Control

After the First Frame or the last Consecutive Frame of a block of frames (when further Consecutive Frames are required to complete the data transfer) the receiver must send a Flow Control (FC) frame. The PCI Type 0011B identifies a Flow Control Frame.
The flow control mechanism allows the receiver to inform the sender about the receiver's capabilities. The Flow Control contains information regarding Block Size and the Separation Time. The Block Size (BS) is the maximum number of Consecutive Frames the receiver allows the sender to send, before waiting for an authorisation (Additional Flow Control Frames) from the receiver to continue transmission. The Separation Time (Stmin) is the (minimum) time the sender has to wait between the transmissions of two consecutive frames.

Flow Status (FS):

Hex Value	Description
0h	Continue To Send (CTS) The Receiver is ready and is waiting for maximally BS Consecutive Frames

Block Size (Bsmax):

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Hex Value	Description
00h	Block Size (BS) Indicates that no more Flow Control (FC) frames are allowed during transmission of the segmented data frame. All Consecutive Frames (CF) will be send without any stop for further Flow Control frames.
01h – FFh	Block Size (BS) Defines the maximal amount of Consecutive Frames (CF) that can be received without an intermediate Flow Control (FC) frame.

Separation Time (Stmin):

Hex Value	Description
00h – FFh	Separation Time (Stmin) The Separation Time defines the minimum allowed time gap (in ms) between the transmission of Consecutive Frames (CF).

7.1.3 CAN Communication Failures

Unexpected Messages

As a general rule, the arrival of an unexpected message or a message with illegal format from any node will be ignored.

Missing Messages

If the ECU is detecting a communication failure (e.g. timing violations of the Network Layer) during a diagnostic tool request, the ECU discards the request and no response will be sent.
If the ECU is not in the Default Session and not receiving any request from the diagnostic tool for S3_ECU the ECU will start the Default Session

7.1.4 CAN Identifier

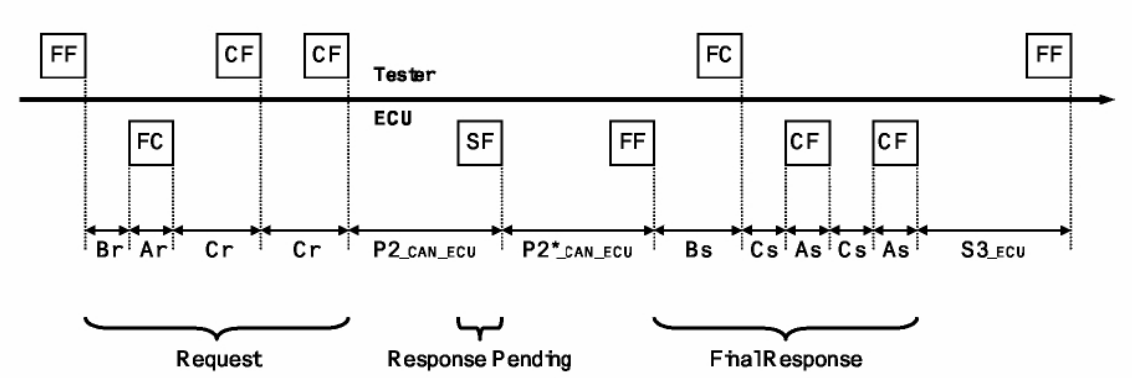
Description	Name	CAN–ID
Physical addressed request messages (customer diagnosis)	CANID_REQ_PHYS	7E1h
Response messages (customer diagnosis)	CANID_RES_PHYS	7E9h
Functional addressed request messages	CANID_REQ_FUNC	7DFh

7.1.5 CAN Timing Parameters

Requirements:

- ECU timings have an accuracy error < - 2 ms

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Timeout Parameters

Parameter	Name	Performance requirement
Time until CAN message has to be transmitted	N_As/N_Ar	1000 ms
Time for transmitting of the next flow control	N_Br	10ms
Time for reception of the next flow control	N_Bs	1000ms
Time until transmission of the next consecutive frame	N_Cs	10ms
Time until reception of the next consecutive frame	N_Cr	1000ms
P2Timeout	P2	50ms
P2*Timeout	P2*	5000ms

7.1.6 CAN Format

Padding

Padding is used when transmitting with fixed DLC of 8. The unused bytes are filled with a defined pattern.

Parameter	Description	Setting
Padding	Fill the unused bytes with a predefined value.	ON
Padding Value	The value which is getting written in unused data bytes	0x00

Buffersize

The Bootloader shall allow the maximal number of bytes which are possible with ISO TP (4095 bytes)

Addressing

According to ISO 15765-2 two different addressing types are defined: Standard / Extended
In Standard addressing all bytes in the can frames are used as databytes (excluding the PCI byte)
In extended addressing the first byte of each frame is used as an extra address information.
For physical CAN ID the TP shall support standard addressing.
For functional CAN ID the TP shall support standard addressing.

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Flow Control Parameter

Parameter	Description	Setting
WFTmax	No of FlowControl wait frames are allowed until a Flow Control CTS has to be received	0
BlockSize	Defines the maximal amount of Consecutive Frames (CF) that can be received without an intermediate Flow Control (FC) frame. Special case 0: no further FC_CTS required.	0

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